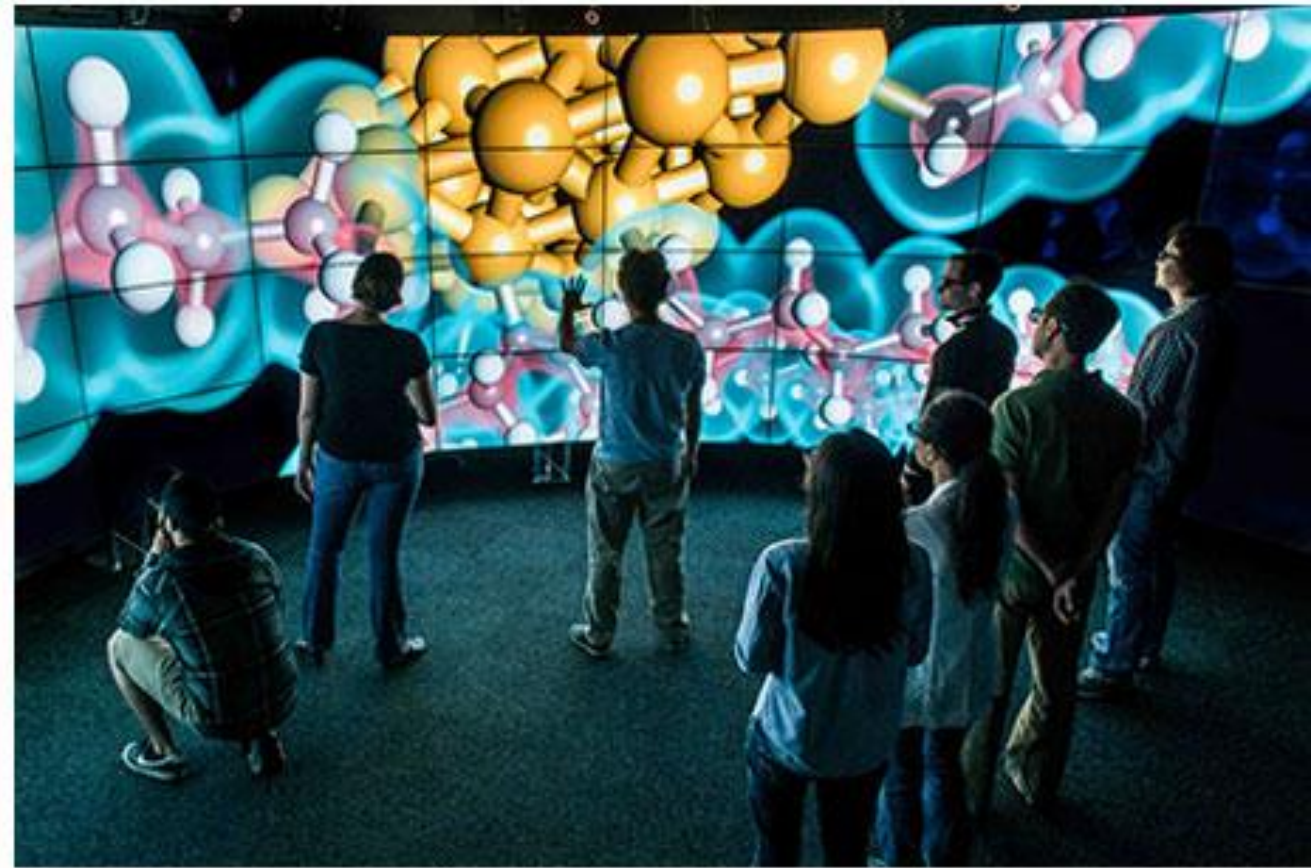




# Chemical Engineering



Industrial Advisory Board Meeting  
July 22<sup>nd</sup> , 2019



# Welcome Advisory Board Members!

## INDUSTRIAL ADVISORY BOARD MEETING AGENDA

- |                      |                                                            |
|----------------------|------------------------------------------------------------|
| <b>9:00-9:30</b>     | Arrival and Check-in                                       |
| <b>9:30-9:45</b>     | Welcome by Professor Zdunek                                |
| <b>9:45-10:30</b>    | State of the Department presentation Professor Berry       |
| <b>10:30-11:00</b>   | ABET update and program objectives discussion              |
| <b>11:00-11:15</b>   | <b>Break</b>                                               |
| <b>11:15 - 12:15</b> | Discussion of undergraduate curriculum changes             |
| <b>12:15 – 12:30</b> | Department Needs                                           |
| <b>12:15 – 12:30</b> | <b>Wrap-up</b>                                             |
| <b>12:30-1:30</b>    | Luncheon                                                   |
| <b>1:30-2:00</b>     | Break before Engineering Innovation Building Celebration   |
| <b>2:00 -</b>        | Engineering Innovation Building Ribbon Cutting Celebration |



# State of the Department





# Upcoming ABET Assessment: Fall 2020

Undergraduate assessment by the Accreditation Board for Engineering and Technology (ABET). Every 6 years; last assessment Fall 2014

ABET Criterion forming the basis of the assessment:

Criterion 1: Students

**Criterion 2: Program Educational Objectives**

**Criterion 3: Student Outcomes**

Criterion 4: Continuous Improvement

Criterion 5: Curriculum

Criterion 6: Faculty

Criterion 7: Facilities

Criterion 8: Institutional Support

**ABET Self-Study Report: Summer 2019 ABET Review: Fall 2019**



# **ABET Assessment: Fall 2020**

## **Overview of Ongoing Tasks:**

- (1) Organizing assessment data in COE digital portal; will provide easily accessible webpage format for ABET reviewers, better efficiency in data collection and archiving (less paper).
- (2) CHE Portal: Consolidating past and current assessment activities, curriculum data and analysis through May 2019
- (3) Organizing assessment data in relation to the Criteria in preparation for the self-study report during upcoming academic year 2019-2020.



# **ABET Assessment Progress:**

## **Program Educational Objectives:**

- To be Reviewed and Approved by faculty and IAB (today)

## **Student Learning Outcomes:**

- Reviewed and Mapped to CHE courses by faculty instructors
- List of Performance Indicators and course assessment (ongoing)

## **Program Curriculum:**

- Faculty review of CHE courses: Curriculum Day Fall 2019
- Curriculum change proposal (To be discussed today)





## **ABET Assessment Progress:**

### **Course Portfolios:**

- Consolidation and archiving of course outputs in portal (ongoing)

### **Course Self-Improvement:**

- Analysis of student course surveys, alumni surveys, IAB recommendations (ongoing)

### **Senior Exit Survey and Exam:**

- Analysis and assessment (ongoing)

### **Engineering Ethics:**

- *ABET emphasizing Ethics component in curriculum*
- Assessing our coverage of Ethics in ChE courses (Senior Design, Professional Development Seminar course, 200- and 300 level courses)



# **Review & IAB Approval of Current Program Educational Objectives:**

| PEO | Category           | Description                                                                                                                                                                                                                                                                                                  |
|-----|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Depth              | In engineering practice, advanced study or original research, UIC Chemical Engineering bachelor's degree graduates will be effective in applying fundamental principles, scientific knowledge, rigorous analysis, creative design, and conceptual innovation.                                                |
| 2   | Breadth            | Based upon mastery of engineering in a broad, societal context, UIC Chemical Engineering graduates will have successful careers in the public or private sectors, or in the pursuit of graduate education.                                                                                                   |
| 3   | Professionalism    | In the service of industry, government, the engineering profession and society at large, graduates will function effectively in the complex modern work environment with clear communication, responsible teamwork, and high standards of ethics, professionalism, safety and protection of the environment. |
| 4   | Life-long learning | Based upon a rigorous undergraduate program that is innovative, challenging, open and supportive, graduates will enhance their skills and knowledge through life-long learning and demonstrate professional leadership.                                                                                      |





## Mapping of ABET Student Learning Outcomes: Old 2014 to New 2020 ABET Outcomes

| New Student Learning Outcomes ( For ABET 2020)                                                                                                                                                                                                  | Old Student Learning Outcomes (ABET 2014)                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.                                                                                                  | (1) Ability to apply knowledge of mathematics, science and engineering<br>(2) Ability to identify, formulate and solve engineering problems                                                                                                                                                                                                       |
| (2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.                   | (3) Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice<br>(7) Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability |
| (3) an ability to communicate effectively with a range of audiences.                                                                                                                                                                            | (12) Ability to communicate effectively                                                                                                                                                                                                                                                                                                           |
| (4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts. | (8) Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context<br>(9) Knowledge of contemporary issues<br>(13) Understanding of professional and ethical responsibilities                                                                                             |
| (5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.                                                 | (11) Ability to function on multidisciplinary teams                                                                                                                                                                                                                                                                                               |
| (6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.                                                                                                | (4) Ability to design experiments<br>(5) Ability to conduct experiments<br>(6) Ability to analyze and interpret data                                                                                                                                                                                                                              |
| (7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.                                                                                                                                             | (10) Recognition of the need for and ability to engage in life-long learning                                                                                                                                                                                                                                                                      |



## ABET Year 2014 Student Outcomes

| SO | Description                                                                                                                                                                                                                |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Ability to apply knowledge of mathematics, science and engineering                                                                                                                                                         |
| 2  | Ability to identify, formulate and solve engineering problems                                                                                                                                                              |
| 3  | Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice                                                                                                                |
| 4  | Ability to design experiments                                                                                                                                                                                              |
| 5  | Ability to conduct experiments                                                                                                                                                                                             |
| 6  | Ability to analyze and interpret data                                                                                                                                                                                      |
| 7  | Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability |
| 8  | Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context                                                                                     |
| 9  | Knowledge of contemporary issues                                                                                                                                                                                           |
| 10 | Recognition of the need for and ability to engage in life-long learning                                                                                                                                                    |
| 11 | Ability to function on multidisciplinary teams                                                                                                                                                                             |
| 12 | Ability to communicate effectively                                                                                                                                                                                         |
| 13 | Understanding of professional and ethical responsibilities                                                                                                                                                                 |



# Break





# Undergraduate Curriculum Changes



# Current Chemical Engineering Curriculum

## Undergraduate Core in ChE

### Thermodynamics-Based Courses

ChE 201 – Introduction to thermodynamics  
ChE 301 – Chemical Engineering Thermodynamics  
ChE 321 – Chemical Reaction Engineering

### Transport-Based Courses

ChE 311 – Transport Phenomena I (Fluid Mechanics)  
ChE 312 – Transport Phenomena II (Heat & Mass Transfer)  
ChE 313 – Transport Phenomena III (separations)

### Process-Oriented Courses

ChE 210 – Material & Energy Balances  
ChE 341 – Chemical Process Control  
ChE 396 – Senior Design I (Equipment Design)  
ChE 397 – Senior Design II (Design teams)

### Unit Operations Laboratory

ChE 381 – Chem Eng Lab I  
ChE 382 – Chem Eng Lab II

### Numerical Analysis

ChE 205- Computational Methods in ChE

### Engineering Education & Development

ENGR100 -Engineering Orientation (Freshmen Success Program, Guaranteed Paid Internship Program)  
ChE 499- Professional Development Seminar



## Technical Electives in ChE (One Required for BS degree)

CHE 392-Undergraduate Research

**CHE 410-Transport Phenomena**

CHE 422-Biochemical Engineering

CHE 425-Nanotechnology for  
Pharmaceutical Applications

**CHE 431 Numerical Methods in ChE**

CHE 433-Process Simulation  
With Aspen Plus

CHE 438-Computational Molecular Modeling

CHE 440-Non-Newtonian Fluids

**CHE 445-Mathematical Methods  
In Chemical Engineering**

CHE 450-Air Pollution Engineering

ChE 451-Renewable Energy Technologies

**ChE 494-Electrochemistry → CHE 451**

ChE 494-Membrane Technology

CHE 494 Fizzics/Interfacial Phenomena

CHE 494 Entrepreneurship in  
Chemical Engineering

CHE 494 2-D Nanomaterials

CHE 494 Biotechnology and  
Nanobiotechnology

CHE 494 Solar Fuel Engineering

CHE 494 Polymer Processing and  
Properties





# Other Courses in ChE Curriculum

## Science/Math and Engineering Courses

CS 109 -C/C++ Programming w/MatLab (**Note: CS Dept is modifying this course**)

ECE 210- Electrical Circuit Analysis

**CME 260- Property of Materials**

MATH 180, 181, 210, 220- Calculus I, II, III and Differential Equations

Physics 141, 142- General Physics I and II

CHEM 122/123, 124/125 General Chemistry I and II with Lab

CHEM 222 Analytical Chemistry

CHEM 232, 233, 234 Organic Chemistry I and II, Organic Chemistry Lab I

CHEM 342 Physical Chemistry I

**CHEM 346 Physical Chemistry II**

**Other Courses:** English Composition I and II, Six General Education Courses



## BS Degree Curriculum Change Proposal

- Adding five concentrations to BS-degree
  - Currently have Biochemical Engineering Concentration
- Adding five new courses
- Changing course requirements
- Converting three, 400-level courses to 500-level (graduate level)



## Introduction of New Concentrations

- Concentration in Biochemical Engineering (Currently Offered)
- Concentration in Process Simulations
- Concentration in Polymer and Molecular Engineering
- Concentration in Nanotechnology
- Concentration in Energy and Environment
- Concentration in Intellectual Property and Entrepreneurship





## Introduction of New CHE Courses

- CHE 2XX: Molecular Systems in Chemical Engineering **(Core Course)**
- CHE 3XX: Polymer Science **(Core Course)**
  - CHE301 and CHE311 as prerequisites
- CHE 4X1: Molecular and Macromolecular Engineering **(Elective)**
- CHE 4X2: Nanoscale Systems in Chemical Engineering **(Elective)**
- CHE 4X3: Colloidal and Interfacial Phenomena **(Elective)**



## Change in Course Requirements

- Choice between CHE 3XX (Polymer Science) and CHEM 346 (Physical Chemistry - II) as a required course for Bachelor of Science in Chemical Engineering degree.
- Choice between CHE 2XX and CME 260 towards BS degree

## **CONCENTRATION IN PROCESS SIMULATIONS**

### **1. REQUIRED COURSES**

- a. CHE 2XX (between CHE 2XX and CME260)
- b. CHE 3XX (Core Class to be taken by all anyways)
- c. CHE 433. Process Simulation with Aspen Plus (as Technical Elective)

### **2. ELECTIVE COURSES**

- a. Elective from outside ChE-rubric from the following list:
  - i. ECON 120 and ECON 220 Theory and Applications of Microeconomics (+3 credits)
  - ii. IE 201 Financial Engineering
  - iii. MATH 310 Applied Linear Algebra
  - iv. MATH 419 Models in Applied Mathematics

TOTAL ADDITIONAL CREDITS FOR THE CONCENTRATION = 3 credit hours, if ECON 120/220 is taken; otherwise 0 additional credits



## CONCENTRATION IN POLYMER AND MOLECULAR ENGINEERING

### 1. REQUIRED COURSES

- a. CHE 2XX (between CHE 2XX and CME260)
- b. CHE 3XX Polymer Science
- c. CHE 4X1 Molecular and Macromolecular Engineering (as Technical Elective).

### 2. ELECTIVE COURSES

- a. Elective from outside ChE-rubric from the following list:
  - i. PHYS 450/BIOE 450. Molecular Biophysics of the Cell
  - ii. PHYS 461 Thermal and Statistical Physics
  - iii. BIOE 405. Atomic and Molecular Nanotechnology
  - iv. CHEM 458. Biotechnology and Drug Discovery
  - v. CHEM 448 Statistical Thermodynamics
  - vi. CHEM 452/454 Biochemistry
  - vii. PHAR 422. Fundamentals of Drug Action.
  - viii. PHAR 423. Biomedical Chemistry
- b. Second Elective from the following list:
  - i. CHE 4X2 Nanoscale Systems in Chemical Engineering
  - ii. CHE 4X3: Colloidal and Interfacial Phenomena
  - iii. CHE 494: 2D Nanomaterials
  - iv. CHE 494: Electrochemistry
  - v. CHE 425. Nanotechnology for Pharmaceutical Applications
  - vi. CHE 494: Macromolecular and Supramolecular Synthesis
  - vii. CHE 438: Computational Molecular Modeling
  - viii. CHE 494: Soft-Matter and Complex Fluids
  - ix. CHE 440: Non-Newtonian Fluids
  - x. CHE 494: Nano and Micro Engineered Membranes
  - xi. CHE 392: Undergraduate Research on Topics of Polymer and Molecular or Engineering

TOTAL ADDITIONAL CREDITS FOR THE CONCENTRATION = 3

## **CONCENTRATION IN NANOTECHNOLOGY**

### **1. REQUIRED COURSES**

- a. CHE 2XX Molecular Systems in Chemical Engineering (between CHE 2XX and CME260)
- b. CHE 3XX: Polymer Science (Core Course)
- c. CHE 4X2 Nanoscale Systems in Chemical Engineering (as Technical Elective)

### **2. ELECTIVE COURSES**

- a. Elective from outside ChE-rubric from the following list:
  - i. ECE 346. Solid State Device Theory.
  - ii. ECE 440. Nanoelectronics.
  - iii. ECE 449. Microdevices and Micromachining Technology.
  - iv. BIOE 405. Atomic and Molecular Nanotechnology
  - v. BIOE 485. Nanobiosensors
  - vi. CHEM 458. Biotechnology and Drug Discovery
  - vii. ME 418. Transport Phenomena in Nanotechnology.
  - viii. PHYS 431: Modern Physics: Condensed Matter.
- b. Second Elective from the following list:
  - i. CHE 494: 2D Nanomaterials
  - ii. CHE 494: Nanomaterials for Energy Applications
  - iii. CHE 494: Colloidal and Interfacial Phenomena
  - iv. CHE 425. Nanotechnology for Pharmaceutical Applications.
  - v. CHE 494 (With Topics related to Nanotechnology)
  - vi. CHE 392: Undergraduate Research on Topics Thematically linked to Nanotechnology

**TOTAL ADDITIONAL CREDITS FOR THE CONCENTRATION = 3**

## **CONCENTRATION IN ENERGY AND ENVIRONMENT**

### **1. REQUIRED COURSES**

- a. CHE 2XX (between CHE 2XX and CME260)
- b. CHE 3XX (Core Class to be taken by all anyways)
- c. CHE 451. Renewable Energy Technologies (Technical Elective)
- d. CME 322. Environmental Engineering. (Additional)
- e. CME 421. Water Treatment Design (Elective from outside ChE-rubric).

### **2. ELECTIVE COURSES**

- a. Second Elective from the following list:
  - i. CHE 494: Electrochemistry
  - ii. CHE 494: Nano and Micro Engineered Membranes
  - iii. CHE 392: Undergraduate Research on Topics Thematically linked to Sustainability, Electrochemistry, Water Science, and Renewable Resources.
  - iv. CHE 450. Air Pollution Engineering

**TOTAL ADDITIONAL CREDITS FOR THE CONCENTRATION = 6**

## **CONCENTRATION IN ENTREPRENEURSHIP AND INTELLECTUAL PROPERTY**

### **1. REQUIRED COURSES**

- a. CHE 2XX (between CHE 2XX and CME260)
- b. CHE 3XX (Core Class to be taken by all anyways)
- c. CHE 494. Entrepreneurship in Chemical Engineering (as Technical Elective)

### **2. ELECTIVE COURSES**

- a. Elective from outside ChE-rubric from the following list:
  - i. ENGR 401. Engineering Management
  - ii. ENGR 402. Intellectual Property Law
  - iii. ENGR 404. Entrepreneurship.

**TOTAL ADDITIONAL CREDITS FOR THE CONCENTRATION = 0**





# Alumni Survey Results on Program Changes

- 28% Response (96/340)
- First Cut: Most Important Concentration by Category (Chemical Industry, Other Industries, Emerging Industries, Academia)

# Chemical Industry

Energy and Environment

Process Simulation

Polymer and Molecular

Intellectual Property/  
Entrepreneurship

Nanotechnology

| Energy and Env.. |    |
|------------------|----|
| 2                | 1  |
| 3                | 1  |
| 5                | 1  |
| 6                | 4  |
| 7                | 6  |
| 8                | 9  |
| 9                | 9  |
| Most Important   | 16 |

| Process Simula.. |    |
|------------------|----|
| 5                | 2  |
| 6                | 2  |
| 7                | 9  |
| 8                | 7  |
| 9                | 14 |
| Most Important   | 13 |

| Polymer and M.. |    |
|-----------------|----|
| 4               | 1  |
| 5               | 10 |
| 6               | 1  |
| 7               | 9  |
| 8               | 10 |
| 9               | 8  |
| Least Important | 1  |
| Most Important  | 7  |

| Intellectual Pro.. |   |
|--------------------|---|
| 2                  | 2 |
| 3                  | 3 |
| 4                  | 4 |
| 5                  | 7 |
| 6                  | 9 |
| 7                  | 7 |
| 8                  | 4 |
| 9                  | 3 |
| Least Important    | 2 |
| Most Important     | 6 |

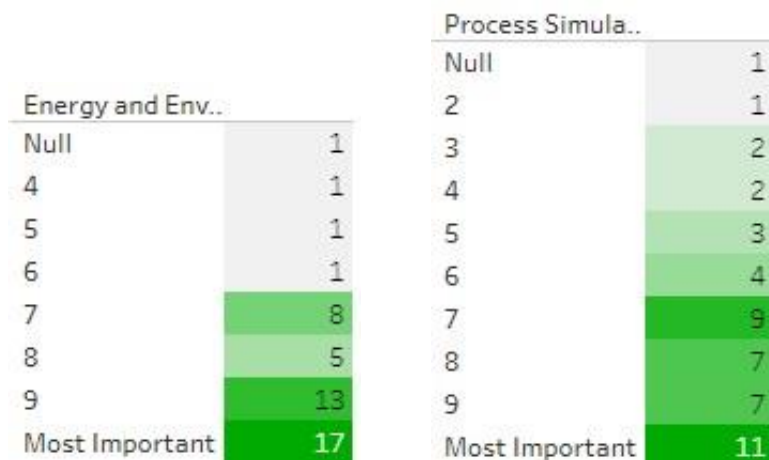
| Nanotechnolgy  |   |
|----------------|---|
| 2              | 1 |
| 3              | 6 |
| 4              | 2 |
| 5              | 8 |
| 6              | 6 |
| 7              | 7 |
| 8              | 7 |
| 9              | 6 |
| Most Important | 4 |

Highest  
“MOST  
IMPORTANT”  
Response

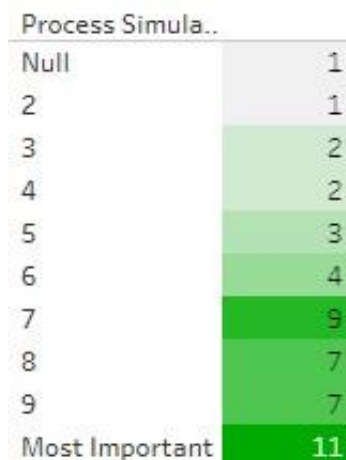
Lowest  
“MOST  
IMPORTANT”  
Response

# Other Industries Hiring ChE's

Energy and Environment



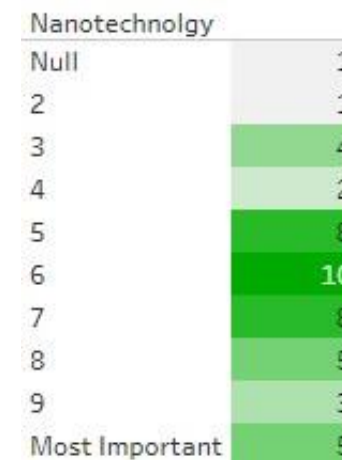
Process Simulation



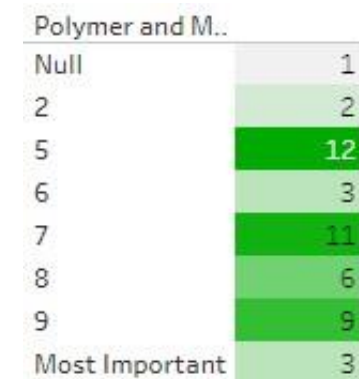
Intellectual Property/ Entrepreneurship



Nanotechnology



Polymer and Molecular

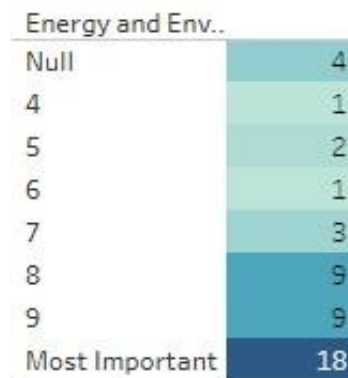


Highest  
"MOST  
IMPORTANT"  
Response

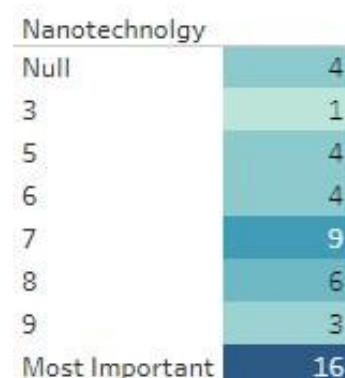
Lowest  
"MOST  
IMPORTANT"  
Response

# Emerging Industries

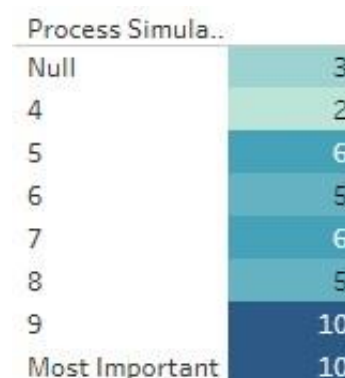
Energy and Environment



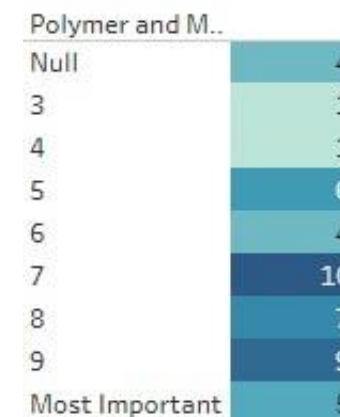
Nanotechnology



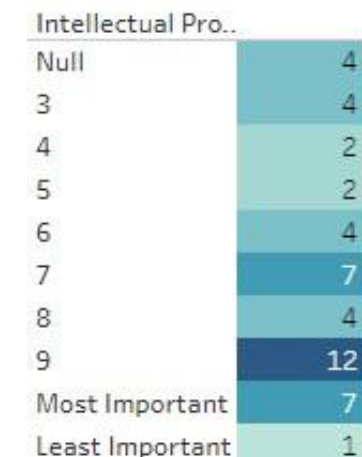
Process Simulation



Polymer and Molecular



Intellectual Property/  
Entrepreneurship



Highest  
"MOST  
IMPORTANT"  
Response

Lowest  
"MOST  
IMPORTANT"  
Response



# Academia

Energy and Environment

| Energy and Env.. |    |
|------------------|----|
| Null             | 5  |
| 3                | 1  |
| 4                | 1  |
| 5                | 2  |
| 6                | 5  |
| 7                | 3  |
| 8                | 5  |
| 9                | 9  |
| Most Important   | 15 |
| Least Important  | 1  |

Nanotechnology

| Nanotechnology |    |
|----------------|----|
| Null           | 5  |
| 3              | 1  |
| 4              | 2  |
| 5              | 1  |
| 6              | 3  |
| 7              | 6  |
| 8              | 6  |
| 9              | 9  |
| Most Important | 14 |

Process Simulation

| Process Simula.. |    |
|------------------|----|
| Null             | 6  |
| 2                | 1  |
| 4                | 3  |
| 5                | 2  |
| 6                | 4  |
| 7                | 9  |
| 8                | 3  |
| 9                | 7  |
| Most Important   | 11 |
| Least Important  | 1  |

Polymer and Molecular

| Polymer and M.. |    |
|-----------------|----|
| Null            | 5  |
| 3               | 1  |
| 4               | 1  |
| 5               | 3  |
| 6               | 3  |
| 7               | 4  |
| 8               | 11 |
| 9               | 10 |
| Most Important  | 9  |

Intellectual Property/  
Entrepreneurship

| Intellectual Pro.. |   |
|--------------------|---|
| Null               | 5 |
| 2                  | 3 |
| 3                  | 2 |
| 4                  | 6 |
| 5                  | 7 |
| 6                  | 5 |
| 7                  | 4 |
| 8                  | 4 |
| 9                  | 4 |
| Least Important    | 2 |
| Most Important     | 5 |

Highest  
“MOST  
IMPORTANT”  
Response

Lowest  
“MOST  
IMPORTANT”  
Response

A word cloud featuring various scientific and technological terms. The words are arranged in a horizontal, slightly overlapping manner. The colors used include purple, red, blue, teal, green, and pink. The sizes of the words vary, with 'SCIENCE' and 'TECHNOLOGY' being the largest. Other prominent words include 'BIG DATA', 'FOOD', 'PHARMACEUTICALS', and 'MATERIAL'. Smaller words like 'BIO-SCIENCES', 'COMPLEX ENGINEERING', 'FLUIDS', 'GAS', 'MOLECULAR SIMULATION', 'PETROLEUM', 'MARKETING', 'EXPLORATION', and 'STUDENTS' are also present.

SCIENCE  
TECHNOLOGY  
BIG DATA  
FOOD  
PHARMACEUTICALS  
MATERIAL  
BIO-SCIENCES  
COMPLEX ENGINEERING  
FLUIDS  
GAS  
MOLECULAR SIMULATION  
PETROLEUM  
MARKETING  
EXPLORATION  
STUDENTS



## Convert 400-level Course to 500-level

- CHE 410 (Transport Phenomena) → CHE 5XX (Advanced Transport Phenomena)
- CHE 431 (Numerical Methods in Chemical Engineering) into CHE 5XX
- CHE 445 (Mathematical Methods in Chemical Engineering) → CHE 5XX



## Department Needs

- Summary and discussion





## Wrap-Up

- Summary and discussion
- Action items
- Next meeting: Spring 2020



# Chemical Engineering Industrial Advisory Board

Thank you for your time



## **INDUSTRIAL ADVISORY BOARD MEETING AGENDA**

- |                      |                                                            |
|----------------------|------------------------------------------------------------|
| <b>9:00-9:30</b>     | Arrival and Check-in                                       |
| <b>9:30-9:45</b>     | Welcome by Professor Zdunek                                |
| <b>9:45-10:30</b>    | State of the Department presentation Professor Berry       |
| <b>10:30-11:00</b>   | ABET update and program objectives discussion              |
| <b>11:00-11:15</b>   | <b>Break</b>                                               |
| <b>11:15 - 12:15</b> | Discussion of undergraduate curriculum changes             |
| <b>12:15 – 12:30</b> | Department Needs                                           |
| <b>12:15 – 12:30</b> | <b>Wrap-up</b>                                             |
| <b>12:30-1:30</b>    | Luncheon                                                   |
| <b>1:30-2:00</b>     | Break before Engineering Innovation Building Celebration   |
| <b>2:00 -</b>        | Engineering Innovation Building Ribbon Cutting Celebration |